Empirical Bayes-augmented convex matrix factorization, completion and denoising

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Abstract

Typical matrix factorization and completion models involve non-convex optimization, so that different initialization or small perturbations to the data can give grossly different solutions. An elegant alternative is to estimate an underlying low-rank "clean" data matrix by enforcing it have small nuclear norm (the convex envelope of matrix rank). This convex optimization problem can be efficiently solved using proximal gradient or Franke-Wolfe methods. However, choosing the regularization strength typically requires cross-validation, involving arbitrary choices (e.g., number of folds), randomness, and substantial computation. Here, we propose empirical Bayes approaches to selecting the regularization strengh, using the recently characterized nuclear norm distribution. We consider multiple noise models, including homoskedastic, known heteroskedastic, and a combination thereof. We present results on simulated data, genetic associations, and single-cell splicing patterns.

Contents

1 Introduction 1

2 Model 2
- 2.1 Homoskedastic setup . . . 2
- 2.2 Limitations of full Bayesian inference for the NND . . . 3
- 2.3 A hybrid solution: convex optimization and empirical Bayes . . . 4
- 2.4 Approximating the marginal likelihood with variational inference . . . 4
- 2.5 Optimization . . . 5
- 2.6 Known heteroskedastic noise . . . 6
- 2.7 Unknown homoskedastic plus known heteroskedastic noise . . . 6
- 2.8 Coordinate Ascent Variational Inference (CAVI) . . . 6
- 2.9 Low-Rank-Plus-Isotropic CAVI via the Woodbury Identity . . . 7

3 Empirical Comparison 9
- 3.1 Lambda Selection Study . . . 11

4 Appendix 11
- 4.1 Mean and Covariance . . . 11
- 4.2 Marginal Distribution of the Nuclear Norm . . . 12

1 Introduction

Low-rank matrix factorization is a workhorse of modern data analysis, underpinning methods ranging from recommender systems to single cell genomics. Yet the overwhelming majority of factorization approaches including non-negative matrix factorization, independent component analysis, and sparse Principal Component Analysis (PCA), optimize a non-convex objective over a product of factor matrices. Non-convexity has practical consequences: different random initializations or minor data perturbations can result in methods finding qualitatively different local minima, making results difficult to reproduce and the recovered factor structure ambiguous [1, 2]. Classical PCA is the notable exception, enjoying a globally optimal and unique solution (up to sign flips) via the singular value decomposition, but it cannot handle missing data or heteroskedastic noise, and it provides no internal mechanism for selecting the rank of the solution.

Convex relaxations offer a principled escape from local minima. The nuclear norm, the convex envelope of matrix rank, yields a globally convex objective that implicitly encourages low-rank solutions without factorizing the data matrix [3, 4]. Solving the nuclear-norm-regularized problem produces a unique, reproducible answer regardless of initialization. A practical difficulty, however, is that the strength of the regularization must be chosen by the analyst. Cross-validation (CV) is the standard remedy, but it is computationally expensive and requires arbitrary choices about the CV scheme (e.g., the number of folds and how to choose heldout matrix elements).

We resolve both limitations simultaneously through an empirical Bayes framework. By interpreting the nuclear norm penalty as the negative log-prior of a *nuclear norm* distribution and placing an uninformative Gamma hyperprior on the regularization strength λ, we obtain a Bayesian interpretation in which λ is estimated from the data alongside the latent matrix X. Crucially, a naive maximum a posteriori (MAP) approach fails here due to the Neyman–Scott paradox: collapsing posterior uncertainty into a point estimate causes the effective nuclear norm to be underestimated, biasing λ upward and over-shrinking the signal. We sidestep this by developing a Coordinate Ascent Variational Inference (CAVI) algorithm that maintains explicit posterior uncertainty over X, leading to well-calibrated automatic regularization without any cross-validation.

2 Model

2.1 Homoskedastic setup

For exposition, we first consider a homoskedastic noise model. Assume we observe elements $\Omega \subseteq \{1, \ldots, m\} \times \{1, \ldots, n\}$ of a noisy data matrix Y. We aim to find a "denoised" low-rank matrix X close to Y (e.g., in squared error). The typical approach would be to represent $X = UV^T$ and optimize over U and V, but this is non-convex and requires careful initialization.

Convex optimization formulation. In the convex optimization/compressed sensing literature, a common alternative is to place a convex penalty directly on X that encourages low-rank structure. The natural choice is the nuclear norm $\|X\|_* = \sum_i s_i(X)$, the sum of the singular values (SVs) s of X. The loss is then,

$$\mathcal{L}(X) = \frac{1}{2} \sum_{ij \in \Omega} (Y_{ij} - X_{ij})^2 + \lambda \|X\|_*. \tag{1}$$

In principle λ can be selected by CV, but this is computationally expensive and requires arbitrary choices about the CV scheme (how many folds, how to select heldout entries, what metric to use).

Eq (1) can be efficiently solved using proximal gradient updates. Split (1) into a smooth part $h(X) = \frac{1}{2}\sum_{ij\in\Omega}(Y_{ij} - X_{ij})^2$ and the nonsmooth nuclear-norm term $\lambda\|X\|_*$. The proximal-gradient step is

$$Z^{(t)} = X^{(t)} - \alpha_t \nabla h(X^{(t)}), \qquad X^{(t+1)} = \mathrm{SVT}_{\alpha_t\lambda}\left(Z^{(t)}\right), \tag{2}$$

where SVT_τ soft-thresholds the singular values by τ. The step size α_t is selected by backtracking until $\mathcal{L}(X^{(t+1)}) \leq \mathcal{L}(X^{(t)})$.

Bayesian interpretation. Here, we ask whether Eq 1 can be re-interpreted from an (empirical) Bayesian viewpoint to enable automatic estimation of λ from the data. The squared error loss can be interpreted as the (scaled) log pdf of a Gaussian likelihood,

$$Y_{ij} \mid X_{ij}, \sigma \sim \mathcal{N}(X_{ij}, \sigma^2), \qquad (i,j) \in \Omega, \tag{3}$$

where σ^2 is the noise variance. We consider the nuclear norm penalty as the negative log-pdf of the recently characterized[5] *nuclear norm distribution* (NND),

$$p(X \mid \eta) = \frac{1}{Z(\eta)} \exp\left(-\eta\|X\|_*\right) \tag{4}$$

where the normalizing constant $Z(\eta) = (mn)!C/\eta^{mn}$, with C the volume of the unit nuclear norm ball. Segert and Wycoff [5] proved that setting the NND rate $\eta = \lambda/\sigma$ (and the prior on σ^2 being log-concave in $1/\sigma$) ensures that $P(X, \sigma|Y, \lambda)$ is unimodal.

2.2 Limitations of full Bayesian inference for the NND

Inference challenges. Segert and Wycoff [5] use Markov chain Monte Carlo (MCMC) to sample X, g and λ under this model. While statistically rigorous, the MCMC is computationally painful, requiring full SVDs at each iteration and tedious tuning to obtain reasonable mixing. We also explored a coordinate ascent variational inference (CAVI) approach using a lesser-known bound for the nuclear norm that introduces an auxiliary covariance parameter (see Appendix **??**). Thehe CAVI algorithm is more computationally efficient, but takes us back to a non-convex optimization problem (although arguably with a better behaved optimization landscape than naive matrix factorization, e.g., we are able to show contraction to a single optimum under certain conditions, see Appendix **??**). Even stochastic variational inference offers no panacea: any reasonable variational approximation immediately reintroduces the non-convexity we are trying to avoid.

Prior behavior. Segert and Wycoff [5] showed that $X \sim NND(\eta)$ is equal in distribution to $X = U\mathrm{diag}(s)V$ where U and V are sampled uniformly from the space of orthonormal matrices, and, assuming (without loss of generality) $m \geq n$, the joint density of the non-zero SVs $s_1 > s_2 > \cdots > s_n > 0$ is,

$$p(s_1, \ldots, s_n) \propto \prod_{i=1}^{n} \underbrace{s_i^{m-n} e^{-\eta s_i}}_{\propto \mathrm{Gamma}(m-n+1,\eta)} \cdot \underbrace{\prod_{1\leq i<j\leq n} (s_i^2 - s_j^2)}_{\text{repulsion term}} \tag{5}$$

This distribution on the SVs of X is concerning for $m > n$ since the shape $m - n + 1 > 1$ means that the SVs are pushed *away* from 0. This is analogous to the Lasso, where

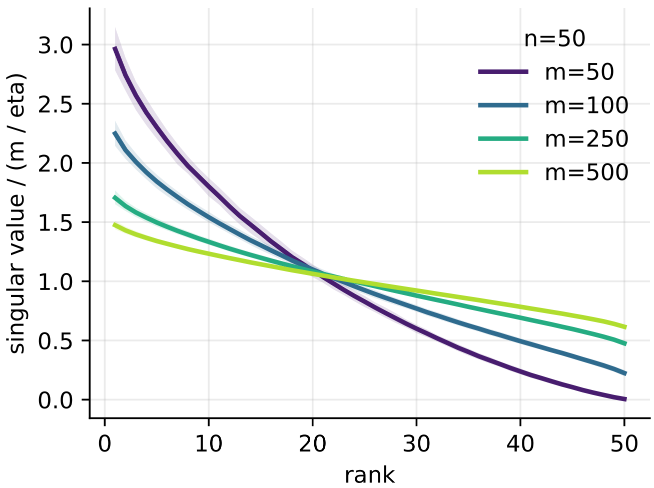

Figure 1: Expected SV profiles under the NND, obtained by NUTS applied to Eq (5). Especially for $m \gg n$, the NND pushes SVs away from zero, which is undesirable for regularization.

optimization under the L1 penalty gives sparse solutions, but Bayesian inference under the analogous Laplace prior does not (motivating spike-and-slab priors, despite their fitting challenges). Note that sparser priors on the SVs would result in non-convex optimization.

2.3 A hybrid solution: convex optimization and empirical Bayes

Given this fundamental mismatch of full Bayesian inference with the NND model we instead explore a hybrid solution. We use the convex optimization machinery of proximal gradient to learn X (MAP) solutions over a grid of η (effective regularization) values. We then derive a closed form variational approximation centered on X via a Taylor series expansion of the expected nuclear norm. This lets us learn the noise variance σ^2 (and implicitly the NND rate $\lambda = \eta\sigma$) via maximum marginal likelihood (i.e. empirical Bayes). The grid of X solutions can then be evaluated by approximate marginal likelihood or leave-one-out (LOO) scores, which are available in closed form given the variational covariance.

We note that we found that joint optimization of the approximate ELBO for λ and σ was unstable.

2.4 Approximating the marginal likelihood with variational inference

We assume a row-factorized Gaussian approximation $q(X) = \prod_{i=1}^{m} \mathcal{N}(x_i \mid \mu_i, \Sigma_i)$ where x_i is row i of X. To cover both the homoscedastic model here and the heteroscedastic models below, we introduce effective noise precisions p_{ij}, with $p_{ij} = 1/\sigma^2$ for $(i, j) \in \Omega$ (for the homoscedastic case) and 0 otherwise. The evidence lower bound (ELBO) is,

$$\mathcal{L}(\mu, \Sigma) = -\frac{1}{2} \sum_{i=1}^{m} \left[(y_i - \mu_i)^\top P_i (y_i - \mu_i) + \mathrm{tr}(P_i \Sigma_i) \right] + \frac{1}{2} \sum_{i=1}^{m} \log |\Sigma_i| + \frac{1}{2} \sum_{(i,j)\in\Omega} \log p_{ij}$$
$$- \eta \mathbb{E}_q \|X\|_* - \log Z(\eta) + \mathrm{const}, \tag{6}$$

where $P_i = \mathrm{diag}\{p_{i:}\}$ is the diagonal precision matrix for row i. The difficulty is $\mathbb{E}_q\|X\|_*$. Write $X = \mu + E$ with $\mathbb{E}_q(E) = 0$ and $\mathbb{E}_q(E^\top E) = \sum_i \Sigma_i$. Using the trace identity $\|X\|_* = \mathrm{tr}\{(X^\top X)^{1/2}\} =: f(X^\top X)$, where

$$X^\top X = \mu^\top \mu + \mu^\top E + E^\top \mu + E^\top E. \tag{7}$$

Expanding f around $\mu^\top \mu$ and taking expectations, the linear terms $\mu^\top E + E^\top \mu$ vanish. Keeping the dominant covariance term gives the matrix-delta approximation

$$\mathbb{E}_q\|X\|_* \approx \|\mu\|_* + \frac{1}{2}\,\mathrm{tr}\left[B(\mu)\sum_{i=1}^{m}\Sigma_i\right]. \tag{8}$$

where $B(\mu) = (\mu^\top \mu + \epsilon I)^{-1/2}$. The small ridge ϵI stabilizes the inverse square root when μ is rank deficient. Substituting (8) into (6), the terms involving a single row covariance are,

$$\mathcal{L}_i(\Sigma_i) = \frac{1}{2}\log|\Sigma_i| - \frac{1}{2}\,\mathrm{tr}\left[\{P_i + \eta B(\mu)\}\Sigma_i\right] + \mathrm{const}. \tag{9}$$

Taking a matrix derivative and setting it to zero gives the optimum,

$$A_i = P_i + \eta B(\mu), \qquad \hat{\Sigma}_i = A_i^{-1}. \tag{10}$$

Because Σ_i has this closed-form optimum given μ, it can be substituted back into the ELBO. The trace term collapses to a constant $-\frac{1}{2}\,\mathrm{tr}(I_n)$ per row, leaving the collapsed ELBO,

$$\begin{aligned}\mathcal{L}_{\mathrm{Taylor}}(\mu) = &-\frac{1}{2}\sum_{(i,j)\in\Omega} p_{ij}(Y_{ij} - \mu_{ij})^2 - \eta\|\mu\|_* - \frac{1}{2}\sum_{i=1}^{m}\log|A_i| \\ &+ \frac{1}{2}\sum_{(i,j)\in\Omega}\log p_{ij} + mn\log\eta + \text{noise-prior terms}.\end{aligned} \tag{11}$$

Here we used that $\log Z(\eta) = -mn\log\eta + \mathrm{const}$.

One might be tempted to fit μ by maximizing (11). However this is 1) computationally challenging at $O(nm^3)$ given the sum of log determinants terms 2) non-convex and 3) statistically ill-advised given the arguments above regarding the downsides of the full Bayesian model.

2.5 Optimization

For the implemented version we use an inverse-gamma variance prior,

$$p(\sigma) \propto \sigma^{-(a_0+1)}\exp(-b_0/\sigma), \tag{12}$$

although the objective separates the prior contribution so that another one-dimensional prior on σ can be substituted. Ideally this should be log-concave in $1/\sigma$ to ensure unimodality of the posterior.

Scoring. Our default strategy is to score solutions using the ELBO (11) evaluated at the proximal mean $\hat{X}$ and the optimal covariance $\hat{\Sigma}_i$ given by (10). However, the same covariance gives an analytical leave-one-entry-out (LOO) score. If $r_{ij} = p_{ij}\Sigma_i[j,j]$, then for observed entries

$$\log p^{\mathrm{LOO}}(Y_{ij}) = -\frac{p_{ij}(Y_{ij} - \hat{X}_{ij})^2}{2(1 - r_{ij})} - \frac{1}{2}\log(2\pi/p_{ij}) + \frac{1}{2}\log(1 - r_{ij}). \tag{13}$$

The LOO score is the sum of these terms over Ω.

2.6 Known heteroskedastic noise

Here, we consider known standard errors $\Delta \in \mathbb{R}^{m \times n}$, where δ_{ij} is the uncertainty associated with Y_{ij},

$$Y_{ij} \mid X_{ij}, \delta_{ij} \sim \mathcal{N}(X_{ij}, \delta_{ij}^2), \qquad (i,j) \in \Omega. \tag{14}$$

In the notation of Eq (6), the precision $p_{ij} = 1/\delta_{ij}^2$ for $(i,j) \in \Omega$ and 0 otherwise. The proximal gradient update is the same as in the homoskedastic case, but with a weighted squared error loss. The ELBO is also the same with the appropriate precision terms. Since the noise is known, there is no noise variance to update, so the only hyperparameter is λ, which is scored by ELBO or LOO over a grid of possible values.

2.7 Unknown homoskedastic plus known heteroskedastic noise

In some settings, we have a combination of known heteroskedastic noise and unknown homoskedastic noise. For example, in single-cell RNA-seq data, we might model the observed counts as subject to both count noise (known heteroskedastic) and other technical noise (unknown homoskedastic). In this setting, the effective precision is $p_{ij} = 1/(\delta_{ij}^2 + \sigma^2)$ for $(i,j) \in \Omega$ and 0 otherwise.

For this variant, $\hat{X} = \mu$ and the noise variance σ^2 are coupled, so we update them by alternating steps. For the mean update we use the proximal gradient step Eq (2), with $\eta = \lambda/\sigma$, where σ^2 is the current noise variance estimate. After the mean update, the scalar variance is updated on the unconstrained scale $\ell = \log \sigma$ by a gradient step on the negative ELBO (11), again with backtracking to enforce monotonicity.

2.8 Coordinate Ascent Variational Inference (CAVI)

Our key insight to enable CAVI is that the nuclear norm can be expressed as a minimization over an auxiliary symmetric positive definite matrix $Q \in \mathbb{S}^n_{++}$,

$$\|X\|_* = \min_{Q \succ 0} \frac{1}{2}\mathrm{Tr}(XQ^{-1}X^T) + \frac{1}{2}\mathrm{Tr}(Q). \tag{15}$$

Augmenting the optimization with Q and applying Jensen's inequality gives a tractable bound on the nuclear norm term of the ELBO:

$$\mathbb{E}_q[-\lambda\|X\|_*] \geq -\frac{1}{2}\lambda\mathrm{Tr}\Big(\mathbb{E}_q[X^T X]\, Q^{-1}\Big) - \frac{1}{2}\lambda\mathrm{Tr}(Q). \tag{16}$$

and naturally results in $q(X)$ factorizing over rows of X. Given the symmetry of the nuclear norm, there is an alternative representation with $Q \in \mathbb{S}^m_{++}$ and $X^T Q^{-1} X$, which would result in a column-wise factorization of $q(X)$. The choice between these is a matter of computational convenience: the row factorization is more efficient when $m \gg n$ and the column factorization is more efficient when $n \gg m$.

Update for $q(X)$. Given Q and λ, the posterior for each row x_i is a multivariate Gaussian, i.e. $q(X) = \prod_{i=1}^m \mathcal{N}(x_i \mid \mu_i, \Sigma_i)$, with mean and covariance given by

$$\Sigma_i = \Big(P_i + \lambda Q^{-1}\Big)^{-1}, \qquad \mu_i = \Sigma_i\,(P_i y_i) \tag{17}$$

The auxiliary matrix Q induces correlation across the columns of X.

Update for Auxiliary Matrix Q. Let $\Psi = \mathbb{E}_q[X^T X] = \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i)$. The value of Q that tightly bounds the expectation is the matrix square root $Q^* = \Psi^{1/2}$.

Update for NND rate λ. In principle we could update λ analytically under a conjugate Gamma prior. However, we generally find is it preferable to treat λ as a hyperparameter and select it by maximizing the ELBO or LOO score over a grid of values.

Batched implementation. For settings where n is moderate (e.g. $n \lesssim 300$) but m is large, the bottleneck of full CAVI is the simultaneous storage of all m row covariances $\Sigma_i \in \mathbb{S}^n_{++}$. We address this using mini-batches of size B. Each batch computes Σ_i via Cholesky, immediately extracts $\mathrm{diag}(\Sigma_i)$ (needed for the ELBO) and the batch's contribution to $\Psi = \sum_i (\mu_i \mu_i^\top + \Sigma_i)$, then discards the full Σ_i before moving to the next batch. Peak memory is $O(Bn^2 + mn)$ rather than $O(mn^2)$.

Rényi α-ELBO for λ selection. We consider the Rényi α-ELBO [6],

$$\mathcal{L}_\alpha = \frac{1}{1-\alpha} \log \int q(X)^\alpha \, p(Y, X \mid \lambda)^{1-\alpha} \, dX, \quad 0 \le \alpha < 1, \tag{18}$$

which satisfies $\mathcal{L}_\alpha \le \log p(Y \mid \lambda)$ and recovers the ELBO as $\alpha \to 1$; at $\alpha = 0$ it is the importance-weighted objective.

For the low-rank-plus-isotropic model with diagonal posterior $q(x_{ij}) = \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$, Gaussian likelihood $p(y_{ij} \mid x_{ij}) = \mathcal{N}(y_{ij}, s_{ij}^2)$, and prior marginal $p(x_{ij}) = \mathcal{N}(0, v_j(\lambda))$, the Rényi integral is a product of Gaussian factors and evaluates in closed form as a correction to the diagonal-Gaussian ELBO:

$$\mathcal{L}_\alpha = \mathcal{L} + \frac{1}{\beta} \sum_{(i,j)\in\Omega} \left[\frac{\beta^2 f_{ij}^2}{2(1 - 2\beta g_{ij})} - \frac{1}{2} \log(1 - 2\beta g_{ij}) \right], \tag{19}$$

where $\beta = 1 - \alpha$,

$$f_{ij} = \sigma_{ij} \left(\frac{y_{ij} - \mu_{ij}}{s_{ij}^2} - \frac{\mu_{ij}}{v_j(\lambda)} \right), \qquad g_{ij} = \tfrac{1}{2} \left(1 - \frac{\sigma_{ij}^2}{s_{ij}^2} - \frac{\sigma_{ij}^2}{v_j(\lambda)} \right), \tag{20}$$

and the sum is over observed entries Ω. The correction vanishes as $\alpha \to 1$ (recovering $\mathcal{L}$) and is negative for $\alpha \in (0, 1)$ (a tighter bound). Crucially, $v_j(\lambda)$ depends on λ through the low-rank-plus-isotropic parameterization, so $\mathcal{L}_\alpha$ is differentiable with respect to λ and can be maximized by gradient-based search (e.g. BFGS on $\log \lambda$). In our implementation (`matlap_iso_renyi_lambda`) we alternate between CAVI updates for the posterior q at fixed λ and a one-dimensional BFGS step to update λ by maximizing (19), providing a fully automatic, cross-validation-free procedure for joint posterior and regularization inference.

2.9 Low-Rank-Plus-Isotropic CAVI via the Woodbury Identity

We avoid $O(mn^3)$ operations by restricting Q to a low-rank-plus-isotropic family,

$$Q = V \, \mathrm{diag}(d) \, V^\top + \delta \, (I - VV^\top), \tag{21}$$

where $\delta > 0$, $V \in \mathbb{R}^{n \times r}$ and $d \in \mathbb{R}^r$ are additional variational parameters.

Update for δ. With $Q^{-1} = V\,\mathrm{diag}(1/d)\,V^\top + \delta^{-1}(I - VV^\top)$, the nuclear-norm ELBO bound decomposes into independent in-subspace and off-subspace terms. The off-subspace contribution $-\frac{\bar{\lambda}}{2}[\mathrm{Tr}(\Psi_\perp)/\delta + (n-r)\delta]$ is minimised in closed form:

$$\delta^* = \sqrt{\frac{\mathrm{Tr}(\Psi_\perp)}{n-r}}, \qquad \mathrm{Tr}(\Psi_\perp) = \mathrm{Tr}(\Psi) - \mathrm{Tr}(\Psi_r), \tag{22}$$

where $\mathrm{Tr}(\Psi) = \sum_i(\|\mu_i\|^2 + \mathrm{Tr}(\Sigma_i))$ and $\mathrm{Tr}(\Psi_r) = \sum_k d_k^2$. This is the exact analogue of $d_k^* = \sqrt{[\Psi]_{kk}}$ for the in-subspace modes: δ^* is the RMS off-subspace eigenvalue of Ψ.

Posterior updates. We set $\gamma = \bar{\lambda}$ as the off-subspace prior precision—identical to what a plain isotropic Gaussian prior $p(x_i) = \mathcal{N}(0, \bar{\lambda}^{-1}I)$ would assign—decoupling the row-posterior update from δ. This prevents the self-reinforcing instability that arises when $\gamma = \bar{\lambda}/\delta$: small δ gives large γ and hence large $\mathrm{Tr}(\Psi_\perp)$, which in turn increases δ^* and weakens the prior. With $\gamma = \bar{\lambda}$ fixed, $\tilde{p}_{ij} = p_{ij} + \bar{\lambda}$ and $c_k = \bar{\lambda}/d_{r,k} - \bar{\lambda} = \bar{\lambda}(1/d_{r,k} - 1)$. Since the in-subspace eigenvalues $d_{r,k}$ are typically large ($d_{r,k} \gg 1$), we have $c_k < 0$; the posterior precision $\Lambda_i = \mathrm{diag}(\tilde{p}_i) + V\,\mathrm{diag}(c_k)\,V^\top$ is nevertheless PD. The Woodbury identity gives

$$\Sigma_i = \mathrm{diag}(\tilde{p}_i^{-1}) - \mathrm{diag}(\tilde{p}_i^{-1})\,V\,\tilde{B}_i^{-1}\,V^\top\,\mathrm{diag}(\tilde{p}_i^{-1}), \tag{23}$$

$$\tilde{B}_i = \mathrm{diag}(1/c_k) + V^\top\,\mathrm{diag}(\tilde{p}_i^{-1})\,V \;\in \mathbb{R}^{r\times r}, \tag{24}$$

where $\tilde{B}_i$ is symmetric and invertible (since $\Lambda_i \succ 0$ implies $\det(\tilde{B}_i) \neq 0$ via the matrix determinant lemma), but need not be positive definite when $c_k < 0$; in practice we diagonalise $\tilde{B}_i = E\,\mathrm{diag}(\ell)\,E^\top$ via `eigh` rather than Cholesky. The posterior mean $\mu_i = \Sigma_i\,\mathrm{diag}(p_i)\,y_i$ follows the same structure and has components outside $\mathrm{col}(V)$, capturing signal not explained by the current factor subspace. $\mathrm{diag}(\Sigma_i)$ is extracted in $O(nr)$ without forming Σ_i explicitly.

***Q* and δ updates.** $\mathrm{Tr}(\Psi_r)$ and V are updated exactly as in Section **??**. $\mathrm{Tr}(\Psi)$ is accumulated in $O(mn)$ from $\|\mu_i\|^2$ and $\mathrm{diag}(\Sigma_i)$, giving $\mathrm{Tr}(\Psi_\perp)$ and thus $\delta^* = \sqrt{\mathrm{Tr}(\Psi_\perp)/(n-r)}$ in $O(n)$. $\mathrm{Tr}(Q) = \sum_k d_{r,k} + (n-r)\delta^*$. Note that $\gamma = \bar{\lambda}$ requires no separate update: it follows immediately from the current $\bar{\lambda}$.

ELBO. By the matrix determinant lemma applied to $\Lambda_i = \mathrm{diag}(\tilde{p}_i) + V\,\mathrm{diag}(c_k)\,V^\top$,

$$\log|\Sigma_i| = -\sum_{k=1}^{r} \log|c_k| - \log|\tilde{B}_i| - \sum_{j=1}^{n} \log(p_{ij} + \gamma), \tag{25}$$

an $O(n + r^3)$ computation that accounts for the full n-dimensional entropy. Absolute values are required because $c_k < 0$ in the typical case ($\delta < d_k$); since $\Lambda_i \succ 0$, the signs of $\prod_k c_k$ and $\det(\tilde{B}_i)$ agree, so $|\det(\Sigma_i)| = (\prod_j \tilde{p}_j)^{-1} \cdot |\prod_k c_k|^{-1} \cdot |\det(\tilde{B}_i)|^{-1}$ is real and positive.

The full ELBO evaluated at the optimal Q (i.e. after substituting δ^*) is

$$\mathcal{L}_{\text{iso}} = \underbrace{-\frac{1}{2}\sum_{(i,j)\in\Omega}\left[\frac{(y_{ij}-\mu_{ij})^2+\Sigma_i[j,j]}{s_{ij}^2}+\log(2\pi s_{ij}^2)\right]}_{\text{log-likelihood}} \underbrace{-\bar{\lambda}\,\mathrm{Tr}(Q)+mn\,\log\lambda}_{\text{nuclear-norm bound at } Q^*}$$
$$+\underbrace{\frac{1}{2}\sum_{i=1}^{m}[\log|\Sigma_i|+n(1+\log 2\pi)]}_{\text{entropy of } q(X)}, \tag{26}$$

where $\mathrm{Tr}(Q)=\sum_k d_k+(n-r)\delta^*$.

ELBO consistency note. One might instead use a Gaussian off-subspace prior $p(x^\perp)=\mathcal{N}(0,\bar{\lambda}^{-1}I)$, which gives the prior term $-\frac{\bar{\lambda}}{2}\mathrm{Tr}(\Psi_\perp)+\frac{m(n-r)}{2}\log\lambda$. At δ^* this equals $-\frac{\bar{\lambda}}{2}(n-r)\delta^{*2}$, whereas the nuclear-norm bound gives $-\bar{\lambda}(n-r)\delta^*$; the two coincide only when $\delta^*=2$. Using the Gaussian ELBO while updating δ and b_N via the nuclear-norm formulae creates an inconsistency that can break ELBO monotonicity for the δ and λ sub-updates. Equation (26) avoids this by using the nuclear-norm bound uniformly.

3 Empirical Comparison

We benchmark all scalable methods on a synthetic matrix completion problem with m = 10,000 rows, n = 1,000 columns, true rank 15, heteroscedastic Gaussian noise ($\sigma_{ij}\sim$ Uniform(0.5, 1.5)), and 20% of entries held out as a test set. We report test-set RMSE and wall-clock time on a NVIDIA RTX 3090 GPU, averaged over 3 seeds.

Methods. `proximal` and `proximal_cv` solve the nuclear-norm penalised least-squares problem via FISTA (100 iterations); λ is set by a heuristic or 5-fold entry-wise CV respectively. `matlap_faem` and `matlap_gradml` optimise the low-rank factor analysis marginal likelihood via EM and gradient descent (Adam) respectively, with automatic empirical-Bayes λ (100/500 steps). `matlap_lowrank` runs the rank-50 low-rank CAVI with automatic empirical-Bayes λ (50 iterations). `matlap_grid_lowrank` runs the same low-rank CAVI on a 7-point log-spaced λ grid with warm-started path (converged at each point), selecting by ELBO. `matlap_grid_lowrank_iso_elbo` and `matlap_grid_lowrank_iso_renyi` run the low-rank-plus-isotropic CAVI on the same grid, selecting by ELBO or Rényi α-ELBO ($\alpha=0.5$) respectively. `matlap_batched` runs exact full CAVI with batch size 64, automatic λ (converged; $O(n^3)$ per row, so slow at n = 1,000). `vi_diagonal` runs SVI with a fully-factorised Gaussian guide and full SVD nuclear norm (200 steps). `vi_diagonal_approx`, `vi_matrix_factor`, and `vi_row_lowrank` use the rSVD nuclear norm approximation (rank 30) with structured guides (200 steps). `mcmc_mala` runs the proximal MALA sampler (300 warm-up, 400 samples, heuristic fixed λ); `mcmc_gibbs` runs the MALA+MH Gibbs sampler (same budget, λ sampled jointly). All MCMC methods use a cold start ($\hat{X}_0=Y$ with missing entries set to zero).

Results.

Discussion. `matlap_faem` and `matlap_gradml` achieve the lowest RMSE (0.081), with runtimes under 10 s. Both optimise the exact marginal likelihood in the factor subspace rather than variational bounds, enabling reliable automatic λ selection. `matlap_grid_lowrank`

Table 1: Test-set RMSE and GPU runtime on a 10,000 × 1,000 rank-15 matrix, 3 seeds (NVIDIA RTX 3090).

Method	RMSE	Time (s)	Converged	Notes
`matlap_faem`	**0.081**	**3.6**	✓	Best RMSE
`matlap_gradml`	0.081	3.2	✓	
`matlap_grid_lowrank`	0.099	2.4	✓	Fastest converged
`matlap_grid_lowrank_iso_elbo`	0.114	27	–	Iso; 50-iter budget
`matlap_grid_lowrank_iso_renyi`	0.124	27	–	Iso; Rényi scoring
`proximal_cv`	0.105	121	–	80× slower
`proximal`	0.123	23	–	Fixed λ
`matlap_batched`	0.153	184	✓	Exact; slow at $n = 1{,}000$
`matlap_lowrank`	0.257	1.0	✓	Auto-λ diverges
`vi_diagonal`	0.242	42	–	200 steps
`vi_matrix_factor`	0.269	19	–	rSVD approx
`vi_row_lowrank`	0.270	25	–	rSVD approx
`vi_diagonal_approx`	0.397	11	–	rSVD approx
`mcmc_mala`	0.258	250	–	Not converged in $T = 700$
`mcmc_gibbs`	0.258	252	–	Not converged in $T = 700$

achieves the best efficiency trade-off (RMSE 0.099 in 1.5 s) by evaluating a warm-started regularisation path on a shared low-rank basis, avoiding the over-shrinkage of the automatic empirical-Bayes λ in factor space (which inflates the effective regularisation by a factor of $n/r \approx 20$).

The low-rank-plus-isotropic variants (`matlap_grid_lowrank_iso_*`) use the corrected $\gamma = \bar{\lambda}$ prior (Section 2.9) and the correct hybrid ELBO normaliser ($mr\mathbb{E}[\log \lambda]$ for the r-dimensional nuclear-norm subspace, $\frac{m(n-r)}{2}\mathbb{E}[\log \lambda]$ for the Gaussian off-subspace). With the corrected ELBO the grid search selects an intermediate $\lambda \approx 158$ (previously it selected the grid maximum due to the monotonically increasing incorrect ELBO). Neither iso variant converges within the 50-iteration budget used here (the per-iteration cost is $\sim 11\times$ higher than pure lowrank due to the off-subspace update); given the budget constraint, RMSE is 0.114 (ELBO scoring) and 0.124 (Rényi scoring), comparable to `proximal_cv` (0.105). With a longer iteration budget, the iso models are expected to match or improve on `matlap_grid_lowrank` since they are strictly more expressive.

`matlap_batched` converges to the exact full-CAVI solution (RMSE 0.153) but is slow at $n = 1{,}000$ because each row requires an $O(n^3)$ Cholesky; it is intended for settings with moderate n ($\lesssim 300$) where exact row posteriors are needed without the full $O(mn^2)$ memory.

The SVI methods (`vi_diagonal`, `vi_matrix_factor`, `vi_row_lowrank`) have not converged at 200 steps, and the rSVD nuclear-norm approximation introduces additional gradient noise in the `approx` variants.

The MCMC samplers (`mcmc_mala` and `mcmc_gibbs`) obtain RMSE 0.258 after 700 total steps — comparable to a zero-signal predictor — indicating that neither chain mixes adequately at the scale of $mn = 10^7$ parameters within this budget. The 250 s per-run cost reflects $O(mn \min(m, n))$ SVD calls at every step; accelerating MCMC to competitive performance at this scale would require either subsampled SVD proposals or a different parameterisation (e.g. sampling in the low-rank factor subspace directly), and we leave this for future work.

3.1 Lambda Selection Study

We also compared λ-selection rules on a 500×100 synthetic problem under two sweeps: varying true rank $(1, 3, 5, 10, 20, 40)$ and varying SNR $(0.1, 0.3, 1, 3, 10)$, with 3 seeds per setting. Table 2 reports mean test RMSE averaged over each sweep.

Table 2: Mean test RMSE across the rank and SNR sweeps.

Method	Rank sweep	SNR sweep
`proximal_cv`	0.467	0.314
`matlap_grid`	0.615	0.450
`lowrank_cv`	0.600	0.485
`lowrank_grid`	0.637	0.634
`iso_cv`	0.585	0.445
`iso_grid_renyi`	0.585	0.446
`iso_grid_is`	0.587	0.464
`iso_grid_loo`	0.631	0.476

The key pattern is that `iso_grid_renyi` is essentially tied with `iso_cv` across both sweeps, while `iso_grid_is` ($\alpha = 0$) is close but degrades at higher SNR and `iso_grid_loo` is too conservative. The ELBO-based grids (`matlap_grid` and `lowrank_grid`) remain systematically worse than CV.

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4 Appendix

4.1 Mean and Covariance

Because the nuclear norm is absolutely homogeneous ($\|-X\|_* = \|X\|_*$) and the density is symmetric around the origin, the expectation of X is exactly the zero matrix: $\mathbb{E}[X] = 0_{m\times n}$.

Furthermore, because $\|X\|_*$ is invariant under left- and right-orthogonal transformations ($p(UXV^T) = p(X)$ for orthogonal U, V), the entries of X are mutually uncorrelated.

The covariance matrix of $\mathrm{vec}(X)$ is spherical:

$$\mathrm{Cov}(\mathrm{vec}(X)) = \frac{C_{m,n}}{\lambda^2} I_{mn} \tag{27}$$

where $C_{m,n}$ is a dimensional constant.

4.2 Marginal Distribution of the Nuclear Norm

Transforming to generalized polar coordinates where the radius is $R = \|X\|_*$, the volume of the nuclear norm ball grows as R^{mn}. The surface area measure therefore scales as R^{mn-1}. Integrating out the directional components, the marginal density of the nuclear norm itself follows a Gamma distribution,

$$\|X\|_* \sim \mathrm{Gamma}(mn, \lambda). \tag{28}$$

This yields the moments $\mathbb{E}[\|X\|_*] = \frac{mn}{\lambda}$ and $\mathrm{Var}(\|X\|_*) = \frac{mn}{\lambda^2}$.